

Pendule simple. Méthode énergétique

TEM.

syst : masse m

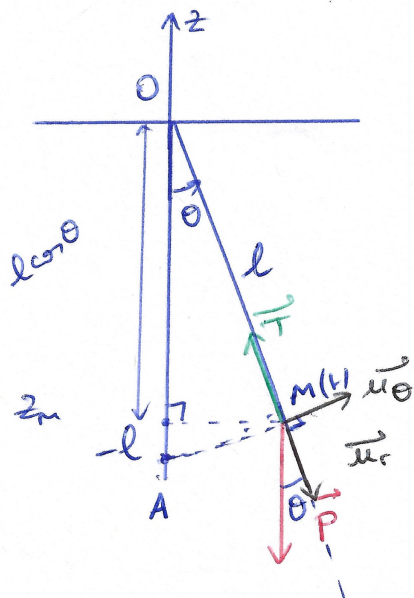
ref : terrestre galiléenne

BDF : - poids $\vec{P}$
- tension $\vec{T}$

$$\sum_i W(\vec{F}_{i \rightarrow c}) = \Delta E_m$$

$$0 = \Delta E_m$$

car $\vec{P}$ conservatif
 $\vec{T}$ ne travaille pas



$$E_{pp}(M(t)) = mgz_m$$

$$= -mgl \cos \theta$$

$$E_c(t) = \frac{1}{2} m l^2 \dot{\theta}^2$$

On choisit $z=0$ pour $E_{pp}=0$

$$\Delta E_m = 0$$

$$z_m = -l \cos \theta(t)$$

On lâche le pendule d'un angle θ_0 avec une vitesse nulle. (part. M_0)

$$E_{pp}(M_0) = -mgl \cos \theta_0$$

$$E_c(M_0) = \frac{1}{2} m v_0^2 = 0$$

$$E_{pp}(M) = -mgl \cos \theta$$

$$E_{pp}(M_0) = -mgl \cos \theta_0$$

$$-mgl(\cos \theta - \cos \theta_0) + \frac{1}{2} m l^2 \dot{\theta}^2 - \frac{1}{2} m v_0^2 = 0$$

$$mgl \sin \theta$$

$$+ \frac{1}{2} m l^2 \ddot{\theta} \dot{\theta}$$

$$= 0 \quad \frac{d}{dt}$$

$$g \sin \theta + l \ddot{\theta} = 0$$

$$\ddot{\theta} + \frac{g}{l} \sin \theta = 0$$

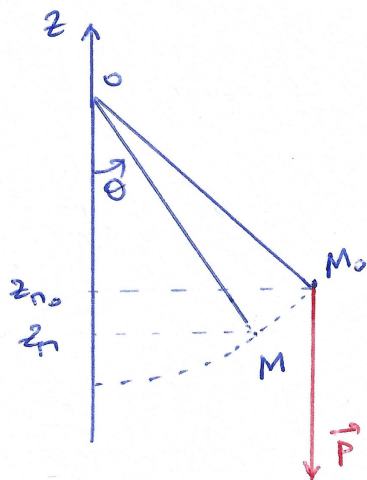
Avec le TEC

$$M_0 \rightarrow P \quad P_0 \rightarrow P$$

$$\Delta E_c = W(\vec{P})$$

$$M_0 \rightarrow M$$

$$W(\vec{P}) = \vec{P} \cdot \vec{M_0 M}$$



$$\vec{OM} = l \sin \theta \vec{u}_r + l \cos \theta \vec{u}_z$$

$$\vec{OM_0} = l \sin \theta_0 \vec{u}_r - l \cos \theta_0 \vec{u}_z$$

$$\vec{M_0 M} = \vec{OM} - \vec{OM_0}$$

$$= \begin{pmatrix} l \sin \theta - l \sin \theta_0 \\ -l \cos \theta + l \cos \theta_0 \end{pmatrix} = l \begin{pmatrix} \sin \theta - \sin \theta_0 \\ -\cos \theta + \cos \theta_0 \end{pmatrix}$$

$$\vec{P} = -mg \vec{u}_z$$

$$\vec{P} \cdot \vec{M_0 M} = mg(\cos \theta - \cos \theta_0)l$$

$$\Delta E_c = -\frac{1}{2}mv_0^2 + \frac{1}{2}mv^2 = mg(\cos \theta - \cos \theta_0)l$$

$$-\frac{1}{2}mv_0^2 + \frac{1}{2}ml^2\dot{\theta}^2 = lmg\cos \theta - lmg\cos \theta_0 \quad \left. \vphantom{\frac{1}{2}ml^2\dot{\theta}^2} \right) \frac{d}{dt}$$

$$+ ml^2\ddot{\theta} = -lmg\sin \theta \cdot \dot{\theta}$$

$$l\ddot{\theta} + g\sin \theta = 0$$

$$\ddot{\theta} + \frac{g}{l}\sin \theta = 0$$